

PROBLEM SET 8

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Physics 16

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Question 1. Simple questions about linearly polarized electromagnetic waves in vacuum, Purcell chapter 9.

- (a) Exercise 9.18 (Associated B Field.) What is the formula for the $\vec{B}$ field if the $\vec{E}$ field of an electromagnetic wave is given by

$$\vec{E} = E_1 \sin \left[\frac{2\pi}{\lambda}(z + ct) \right] (\hat{x} + \hat{y})$$

with $E_1 = 20$ V/m?

Solution. For part (a), let $k = 2\pi/\lambda$. Using Faraday's law,

$$(\nabla \times \vec{E})_x = \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = 0 - E_1 k \cos[k(z + ct)] = -E_1 k \cos[k(z + ct)],$$

$$(\nabla \times \vec{E})_y = \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = E_1 k \cos[k(z + ct)] - 0 = E_1 k \cos[k(z + ct)],$$

$$(\nabla \times \vec{E})_z = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0.$$

Setting $-\partial \vec{B} / \partial t = \nabla \times \vec{E}$ and integrating with respect to t ,

$$B_x = E_1 k \cdot \frac{\sin[k(z + ct)]}{kc} = \frac{E_1}{c} \sin[k(z + ct)],$$

$$B_y = -E_1 k \cdot \frac{\sin[k(z + ct)]}{kc} = -\frac{E_1}{c} \sin[k(z + ct)].$$

Therefore,

$$\boxed{\vec{B} = \frac{20 \text{ V/m}}{3 \times 10^8 \text{ m/s}} \sin \left[\frac{2\pi}{\lambda}(z + ct) \right] (\hat{x} - \hat{y}).}$$

□

- (b) Exercise 9.19 (Find the expressions for a wave.) For a linearly polarized electromagnetic wave with frequency 100 MHz (i.e. 10^8 cycles per second) that is traveling in the $-\hat{x}$ direction and for which the electric field is perpendicular to $\hat{z}$, find expressions for $\vec{E}$ and $\vec{B}$.

Solution. The angular frequency and wave number are

$$\omega = 2\pi f = 2\pi \times 10^8 \text{ rad/s}, \quad k = \frac{\omega}{c} = \frac{2\pi \times 10^8 \text{ rad/s}}{3 \times 10^8 \text{ m/s}} = \frac{2\pi}{3} \text{ rad/m}.$$

Since the wave travels in $-\hat{x}$ and $\vec{E}$ must be perpendicular to both the propagation direction and $\hat{z}$, we have $\vec{E}$ in the $\hat{y}$ direction. For a wave traveling in $-\hat{x}$, the phase is $(kx + \omega t)$, so

$$\vec{E} = E_0 \sin\left(\frac{2\pi}{3}x + 2\pi \times 10^8 t\right) \hat{y},$$

where E_0 is an arbitrary amplitude. Then the magnetic field is $\vec{B} = \frac{1}{c}(-\hat{x}) \times \vec{E}$, which is then

$$\vec{B} = -\frac{E_0}{c} \sin\left(\frac{2\pi}{3}x + 2\pi \times 10^8 t\right) (\hat{x} \times \hat{y}) = -\frac{E_0}{c} \sin\left(\frac{2\pi}{3}x + 2\pi \times 10^8 t\right) \hat{z},$$

$$\vec{B} = -\frac{E_0}{c} \sin\left(\frac{2\pi}{3}x + 2\pi \times 10^8 t\right) \hat{z}.$$

□

Question 2. Purcell chapter 9, exercise 9.23 (Field in a box.) Show that the electromagnetic field described by

$$\vec{E} = E_0 \hat{z} \cos kx \cos ky \cos \omega t,$$

$$\vec{B} = B_0(\hat{x} \cos kx \sin ky - \hat{y} \sin kx \cos ky) \sin \omega t$$

will satisfy the empty-space Maxwell equations Purcell Eq. (9.18), if $E_0 = \sqrt{2} c B_0$ and $\omega = \sqrt{2} c k$. This field can exist inside a square metal box enclosing the region $-\frac{\pi}{2k} < x < \frac{\pi}{2k}$ and $-\frac{\pi}{2k} < y < \frac{\pi}{2k}$, with arbitrary height in the z direction. Roughly what do the electric and magnetic fields look like?

Proof. We NTS that the electromagnetic field obeys all four empty-space Maxwell equations. It is trivial that Gauss' Law of electricity is satisfied since E has no dependence on z . For, Gauss's law for B ,

$$\nabla \cdot \vec{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} = -B_0 k \sin kx \sin ky \sin \omega t + B_0 k \sin kx \sin ky \sin \omega t = 0.$$

Next, for Faraday's Law,

$$\nabla \times \vec{E} = \frac{\partial E_z}{\partial y} \hat{x} - \frac{\partial E_z}{\partial x} \hat{y} + 0 \hat{z} = -E_0 k \cos kx \sin ky \cos \omega t \hat{x} + E_0 k \sin kx \cos ky \cos \omega t \hat{y}.$$

Since

$$-\frac{\partial B_x}{\partial t} = -B_0 \omega \cos kx \sin ky \cos \omega t, \quad -\frac{\partial B_y}{\partial t} = B_0 \omega \sin kx \cos ky \cos \omega t,$$

Faraday's Law is obeyed. Lastly, for Ampère's law,

$$\begin{aligned} (\nabla \times \vec{B})_z &= \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = -B_0 k \cos kx \cos ky \sin \omega t - B_0 k \cos kx \cos ky \sin \omega t \\ &= -2B_0 k \cos kx \cos ky \sin \omega t. \end{aligned}$$

The right-hand side gives

$$\begin{aligned} \frac{1}{c^2} \frac{\partial E_z}{\partial t} &= -\frac{E_0 \omega}{c^2} \cos kx \cos ky \sin \omega t \\ \implies 2B_0 k &= \frac{E_0 \omega}{c^2} \implies \frac{E_0}{B_0} = \frac{2kc^2}{\omega} \\ \implies \frac{\omega}{k} &= \frac{2kc^2}{\omega} \implies \boxed{\omega = \sqrt{2} ck}. \end{aligned}$$

Hence,

$$\frac{E_0}{B_0} = \frac{\omega}{k} = \frac{\sqrt{2} ck}{k} = \sqrt{2} c \implies \boxed{E_0 = \sqrt{2} c B_0}.$$

This field is a standing wave. The electric field oscillates vertically ($\hat{z}$), reaching a maximum at the origin. The magnetic field circulates in the xy -plane, vanishing at the origin and peaking near the walls. The E and B fields are 90 degrees out of phase in time. $\square$

Question 3. Similar to Purcell chapter 9, exercise 9.26 (Wave equation.) Here is a particular electromagnetic field in free space:

$$E_x = 0, \quad E_y = E_0 \sin(kx + \omega t), \quad E_z = 0;$$

$$B_x = 0, \quad B_y = 0, \quad B_z = -\frac{E_0}{c} \sin(kx + \omega t).$$

Show that this field can satisfy the wave equation if ω and k are related in a certain way; what is the relationship between them?

Proof. The wave equation in free space is $\nabla^2 \vec{E} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}$. Applying this to $E_y = E_0 \sin(kx + \omega t)$, the left-hand side is

$$\nabla^2 E_y = \frac{\partial^2 E_y}{\partial x^2} = -k^2 E_0 \sin(kx + \omega t).$$

The right-hand side is

$$\frac{1}{c^2} \frac{\partial^2 E_y}{\partial t^2} = \frac{1}{c^2} (-\omega^2 E_0 \sin(kx + \omega t)).$$

Hence,

$$-k^2 E_0 \sin(kx + \omega t) = -\frac{\omega^2}{c^2} E_0 \sin(kx + \omega t) \implies k^2 = \frac{\omega^2}{c^2} \implies \boxed{c = \frac{\omega}{k}}.$$

□

Question 4. Light from the sun arrives at earth at the rate of $1.4 \times 10^3 \text{ W/m}^2$. For simplicity, assume that sunlight is monochromatic, with a 500 THz frequency. Recall that the energy of a single photon is $E = hf$.

- (a) How many photons fall per second on a square meter of earth's surface perpendicularly facing the sun?
- (b) How many photons per cubic meter are there just above earth?
- (c) What is the total power output of the sun (all directions)? (Hint: Look up the average sun-earth distance.)
- (d) How many photons per second does the sun emit therefore?
- (e) Convert this energy that the sun loses per second via radiation, into an equivalent mass lost per second by using the (in)famous energy-mass relation $E = mc^2$.

Solution. For part (a), the energy of a single photon is

$$E_{\text{photon}} = hf = (6.626 \times 10^{-34} \text{ J} \cdot \text{s})(500 \times 10^{12} \text{ Hz}) = 3.313 \times 10^{-19} \text{ J}.$$

By dimensional analysis, the number of photons per second per square meter is

$$N = \frac{I}{hf} = \frac{1.4 \times 10^3 \text{ W/m}^2}{3.313 \times 10^{-19} \text{ J}} \approx 4.23 \times 10^{21} \text{ photons/(m}^2 \cdot \text{s)}.$$

For part (b), the number density is

$$n = \frac{N}{c} = \frac{4.23 \times 10^{21} \text{ photons/s/m}^2}{3 \times 10^8 \text{ m/s}} \approx 1.41 \times 10^{13} \text{ photons/m}^3.$$

For part (c), we write the intensity inverse square relationship into $I = P_{\text{sun}}/4\pi d^2$, and using $d \approx 1.5 \times 10^{11} \text{ m}$, the total power is

$$P_{\text{sun}} = I \cdot 4\pi d^2 = (1.4 \times 10^3 \text{ W/m}^2) \cdot 4\pi(1.5 \times 10^{11} \text{ m})^2 \approx 3.96 \times 10^{26} \text{ W}.$$

This directly solves part (d) because

$$N_{\text{sun}} = \frac{P_{\text{sun}}}{hf} = \frac{3.96 \times 10^{26} \text{ W}}{3.313 \times 10^{-19} \text{ J}} \approx 1.20 \times 10^{45} \text{ photons/s}.$$

Lastly, for part (e), we use dimensional analysis so that

$$\frac{m}{t} = \frac{P_{\text{sun}}}{c^2} = \frac{3.96 \times 10^{26} \text{ W}}{(3 \times 10^8 \text{ m/s})^2} = \frac{3.96 \times 10^{26}}{9 \times 10^{16}} \approx 4.4 \times 10^9 \text{ kg/s}.$$

The sun loses roughly 4.4 billion kilograms worth of mass-energy per second due to radiation. □